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CASE PORTFOLIO: DATA, MACHINE LEARNING & GENERATIVE AI

2015 ~ 2026

DATA SCIENCE, AI & MLOPS CASES

CASE #1 CORPORATE GENERATIVE AI ASSISTANT WITH ADVANCED RAG

Leading Tech & AI Unicorn

CONTEXT (CHALLENGE)

Implementation of a complex corporate question-answering system based on LLMs, aimed at operational efficiency in technical support.

SITUATION

Technical support teams spent excessive time navigating dense manuals, resulting in high latency for customer service.

TASK

Develop a secure and accurate conversational assistant that reduced resolution times without suffering from technical hallucinations.

ACTION

Architected an advanced RAG (Retrieval-Augmented Generation) pipeline featuring dynamic semantic chunking, refined embeddings, and Cross-Encoder re-ranking integrated into vector databases.

RESULT

Proven 45% reduction in handling time for complex support tickets and a response accuracy rate exceeding 95%.

CASE #2 FINE-TUNING AND EVALUATION OF OPEN-SOURCE LLMS

Leading Tech & AI Unicorn

CONTEXT (CHALLENGE)

Customization of open language models for specific business domains to achieve vendor independence and cost control.

SITUATION

Over-reliance on proprietary APIs (OpenAI) created privacy risks and unsustainable token costs at high scale.

TASK

Fine-tune open-source models (Llama, Mistral) to replicate commercial model performance using internal infrastructure.

ACTION

Executed specialized Fine-Tuning (QLoRA) using sanitized proprietary datasets, applying rigid alignment evaluation frameworks.

RESULT

Internal models operating with performance equivalent to proprietary ones for the business task, achieving a 60% reduction in operational inference costs.

CASE #3 MULTI-MILLION SCALE SEMANTIC SEARCH INFRASTRUCTURE

Leading Tech & AI Unicorn

CONTEXT (CHALLENGE)

Structuring a scalable unstructured data retrieval ecosystem for multiple analytical products across the company.

SITUATION

Traditional keyword-based search engines failed to retrieve the analytical context of millions of legacy documents and logs.

TASK

Establish a high-throughput, low-latency vector database platform.

ACTION

Implemented production vector database clusters (Pinecone/Milvus), designing analytical pipelines for automated embedding generation.

RESULT

Supported complex semantic searches with response times under 50 milliseconds for databases containing hundreds of millions of vectors.

CASE #4 MENTORSHIP AND SCALING IN CORPORATE DEEP LEARNING

Leading Tech & AI Unicorn

CONTEXT (CHALLENGE)

Internal technical training of the analytical team to migrate legacy practices from traditional statistical modeling to Deep Learning.

SITUATION

The data science team was highly focused on traditional linear algorithms, creating an innovation bottleneck for NLP and computer vision problems.

TASK

Elevate the team's analytical maturity and accelerate the adoption of deep neural network techniques.

ACTION

Designed practical, continuous technical mentorship tracks focused on Transformer architectures, PyTorch, and Hugging Face.

RESULT

Upskilled three new analytical squads, enabling them to autonomously deliver deep learning-based solutions.

CASE #5 UNIFIED MLOPS PLATFORM STRUCTURING

Payment Methods Fintech

CONTEXT (CHALLENGE)

Implementation of automated model lifecycle pipelines to eradicate manual deployment processes.

SITUATION

Models built by data scientists remained stuck in silos for weeks due to a lack of delivery standards for software engineering.

TASK

Drastically reduce the time-to-market of the complete model lifecycle (MLOps governance).

ACTION

Designed and orchestrated an integrated platform using MLflow for tracking/reproducibility and Kubernetes for deployment via concurrent FastAPI APIs.

RESULT

Average model validation and deployment time reduced from weeks to under 1 hour.

CASE #6 HIGH-PERFORMANCE REAL-TIME FRAUD DETECTION

Payment Methods Fintech

CONTEXT (CHALLENGE)

Development of predictive mathematical algorithms aimed at mitigating financial transactional risks at the moment of payment.

SITUATION

Organized fraud attacks caused significant operational losses, and the static rule engine caused friction with false positives for legitimate customers.

TASK

Build a fast and assertive predictive model to operate synchronously within the transactional pipeline.

ACTION

Trained advanced tabular neural models combined with XGBoost, utilizing network feature engineering (Graph Networks) and complex statistical data balancing.

RESULT

Immediate reduction of R$ 12 million in annual fraud losses, while maintaining the false positive rate at its lowest historical level.

CASE #7 CONTINUOUS DATA DRIFT AND CONCEPT DRIFT MONITORING

Payment Methods Fintech

CONTEXT (CHALLENGE)

Ensuring continuous statistical stability for critical models in automated credit approval pipelines.

SITUATION

Changes in the macroeconomic environment silently degraded the accuracy of legacy predictive models, driving up delinquency rates.

TASK

Implement an automated continuous monitoring framework to detect anomalies before business performance is impacted.

ACTION

Built Python routines to calculate statistical divergence metrics (such as the Population Stability Index - PSI) daily, comparing production data against the training baseline.

RESULT

Proactive detection of algorithmic degradation, triggering automated retraining and safeguarding the continuous calibration of credit scores.

CASE #8 MASKING AND ANONYMIZATION OF SENSITIVE DATA (LGPD)

Payment Methods Fintech

CONTEXT (CHALLENGE)

Legal compliance adjustment of analytical Data Lakes without destroying the statistical pattern extraction capabilities of Machine Learning solutions.

SITUATION

Data scientists and analysts were directly handling clients' Personally Identifiable Information (PII) in development environments, violating compliance rules.

TASK

Ensure total shielding of the analytical environment in accordance with current data protection laws.

ACTION

Technical coordination of engineering and governance committees to implement automated data encryption and hashing pipelines for sensitive data at ingestion.

RESULT

Total approval in internal compliance audits with zero risk of mapped identity data leaks.

CASE #9 HYPER-PERSONALIZED RECOMMENDATION ENGINE FOR E-COMMERCE

Major Marketplace & Retail Tech

CONTEXT (CHALLENGE)

Development of experience personalization algorithms to increase sales conversion along the consumer journey.

SITUATION

Customers received static and generic product recommendations on the website homepage, leading to low engagement and cold clicks.

TASK

Develop a dynamic recommendation system using collaborative filtering and embeddings that responds to user behavior in real time.

ACTION

Modeled advanced hybrid algorithms fueled by historical and real-time behavioral data (clicks, searches, cart additions).

RESULT

Measurable 18% increase in the platform's Average Order Value (AOV) through automated cross-selling.

CASE #10 MEDALLION ARCHITECTURE (LAKEHOUSE) WITH DATABRICKS AND PYSPARK

Major Marketplace & Retail Tech

CONTEXT (CHALLENGE)

Structuring an elastic central data repository to democratize mass analytical consumption in an organized manner.

SITUATION

Queries took hours due to the explosive growth of unstructured navigation data and the fragmentation of legacy databases.

TASK

Design a robust processing infrastructure capable of ingesting and cleaning dozens of terabytes of data daily.

ACTION

Built a Medallion architecture divided into layers (Bronze, Silver, Gold) using structured parallel PySpark within Databricks.

RESULT

Efficient and secure processing of over 50 TB of daily user behavior logs, reducing corporate reporting times by 90%.

CASE #11 PREDICTIVE MODELING OF LTV AND PURCHASE PROPENSITY

Major Marketplace & Retail Tech

CONTEXT (CHALLENGE)

Using analytical data intelligence to drive smart direct marketing campaigns and optimize media investments.

SITUATION

Push notifications and emails were generic and based on intuition, causing high opt-out rates and wasting marketing budget.

TASK

Mathematically predict the customer's financial Lifetime Value (LTV) and their purchase probability for the upcoming weeks.

ACTION

Applied supervised statistical Machine Learning algorithms (survival models and gradient boosting) to the consolidated historical consumption database.

RESULT

Surgical audience segmentation yielding record ROI on campaigns and a drastic reduction in active user churn.

CASE #12

CONSTRUCTION AND GOVERNANCE OF CENTRAL FEATURE STORES

Major Marketplace & Retail Tech

CONTEXT (CHALLENGE)

Creation of a corporate central repository for pre-computed predictive variables ready for organizational data models.

SITUATION

Different data science squads repeatedly recalculated the same variables (such as average purchases), causing mathematical misalignment and wasting processing power.

TASK

Unify and standardize analytical training and inference features with total consistency.

ACTION

Implemented a unified variable repository (Feature Store) with automated pipelines to deliver data in both batch and low-latency online modes.

RESULT

Complete eradication of code duplication and significant acceleration in the development timeline of new statistical fronts.

CASE #13

STRATEGIC MIGRATION OF BIG DATA TO CLOUD-NATIVE

Analytical Consulting Specialized in IT

CONTEXT (CHALLENGE)

Designing analytical infrastructure transition plans from local physical servers (on-premises) to elastic managed ecosystems in the cloud.

SITUATION

Scalability limitations in legacy local servers made it impossible to process data and advance new complex analytical projects.

TASK

Migrate complex databases and historical data seamlessly, structuring modern scalable architectures on AWS and GCP.

ACTION

Designed and provided technical leadership for migration modeling using modern native solutions such as AWS Glue, Amazon Athena, and Google BigQuery.

RESULT

Migration successfully executed, ensuring analytical stability for reporting and locking operational costs directly to on-demand elastic usage.

CASE #14

MODERN DATA PIPELINES WITH APACHE AIRFLOW AND DBT

Analytical Consulting Specialized in IT

CONTEXT (CHALLENGE)

Implementing modern analytics engineering concepts to transform raw data into quality corporate intelligence.

SITUATION

Multiple fragmented scripts ran in isolation, frequently breaking executive dashboards due to a lack of prior table validation.

TASK

Ensure full control, clear workflow dependencies, and near-real-time reporting supported by rigid validation.

ACTION

Architected centralized DAGs orchestrated via Apache Airflow alongside the strategic use of dbt (data build tool) for data transformation and automated quality testing.

RESULT

Stable and transparent corporate pipelines, securing morning executive reports with validated and sanitized data from end to end.

CASE PORTFOLIO ANALYSIS: PROJECT MANAGEMENT

Case Portfolio Analysis: Project Management

Visualization of the distribution of the 28 cases based on the balance between **Hard Skills** (Technical/Processes) and **Soft Skills** (Leadership/Resilience).

CASE #15

TRANSLATION OF CORPORATE NEEDS (DATA TRANSLATION)

Analytical Consulting Specialized in IT

CONTEXT (CHALLENGE)

Consultative performance to convert operational business pain points into exact mathematical hypotheses addressable by the data team.

SITUATION

A communication gap between product managers (focused on business) and technicians (focused on analytical infrastructure) resulted in data deliverables that lacked real decision value.

TASK

Bridge the analytical gap by designing Proofs of Concept (PoCs) oriented toward actual business ROI.

ACTION

Conducted analytical workshops, mapped revenue pain points, and drafted technical mathematical specifications for viable statistical hypotheses.

RESULT

Guaranteed total strategic alignment, delivering analytical solutions with validated, direct operational revenue gains.

CASE #16

SELF-SERVICE BI TRAINING AND DATA LITERACY

Analytical Consulting Specialized in IT

CONTEXT (CHALLENGE)

Dissemination and democratization of data consumption across operational and managerial layers of corporate clients.

SITUATION

Directors and managers constantly depended on the engineering team for basic spreadsheet data extractions, bottlenecking the technology team's analytical capacity.

TASK

Empower and democratize secure corporate analytical autonomy using modern tools.

ACTION

Designed and delivered practical training programs focused on Self-Service BI tools and fundamental concepts of statistical analysis and interpretation.

RESULT

Drastic reduction in the queue of simple data requests sent to the technical team, establishing a genuine data-driven culture across the company.

CASE #17 ADVANCED CREDIT RISK MODELING (CREDIT SCORING)

Payment Methods Fintech

CONTEXT (CHALLENGE)

Refinement of predictive mathematical algorithms designed to assess the payment capacity and behavior of applicant profiles.

SITUATION

Classic linear statistical models suffered from high delinquency rates under new, dynamic market scenarios.

TASK

Develop a highly precise non-linear predictive model to calibrate the risk appetite of the financial operation.

ACTION

Modeled using complex structured machine learning (Gradient Boosting and Neural Networks), mapping non-traditional behavioral consumption variables.

RESULT

Significant improvement in the model's Gini metric and a sharp drop in the delinquency rate of the active portfolio within the first analysis window.

CASE #18 LARGE-SCALE DEMAND FORECASTING WITH TIME SERIES

Major Marketplace & Retail Tech

CONTEXT (CHALLENGE)

Optimization of supply chains and planning through rigorous mathematical projections based on historical behavior.

SITUATION

Constant stock misalignment caused stockouts of high-demand products and excessive costs from low-turnover items sitting idle in warehouses.

TASK

Predict with mathematical precision the future consumption of thousands of simultaneous SKUs for the following weeks.

ACTION

Designed analytical pipelines using hybrid statistical approaches combining complex autoregressive models and deep networks for time series.

RESULT

Sharp reduction in stockout costs and a major efficiency gain in distributed logistics and warehousing.

CASE #19

DYNAMIC PRODUCT PRICING ENGINE

Major Marketplace & Retail Tech

CONTEXT (CHALLENGE)

Development of elastic algorithms to adjust merchandise values in real time based on stock, competition, and demand.

SITUATION

Manual fixed pricing caused the company to lose profit margins during high-demand peaks or lose sales to competitors during low periods.

TASK

Automate pricing intelligence through a dynamic engine driven by continuous market data.

ACTION

Architected a predictive system that calculated the price elasticity of demand and adjusted values on web platforms in real time.

RESULT

Immediate efficiency gain in aggregate net operational margin and a volumetric increase in monitored daily sales.

CASE #20

BIG DATA SCALE SENTIMENT ANALYSIS

Leading Tech & AI Unicorn

CONTEXT (CHALLENGE)

Natural language processing focused on extracting the true perception of the user base from unstructured media and logs.

SITUATION

Thousands of daily interactions across networks and support channels were ignored in product planning due to the impossibility of manual mass review.

TASK

Proactively categorize and extract semantic value from millions of scattered text strings on a continuous basis.

ACTION

Built distributed processing pipelines feeding specialized NLP models fine-tuned to classify polarities and emotional triggers.

RESULT

Real-time mapping of customer sentiment, providing critical, automated insights directly to product engineering teams.

CASE #21

VECTOR SEMANTIC SEARCH ENGINE FOR CONTENT

Leading Tech & AI Unicorn

CONTEXT (CHALLENGE)

Implementation of new browsing experiences based on the true meaning of user queries, overcoming string syntax restrictions.

SITUATION

Users failed to find relevant content and analytical products if they typed terms with minor typos or synonyms that weren't explicitly mapped.

TASK

Revolutionize the corporate internal search system, delivering results based strictly on latent intent.

ACTION

Used Generative AI bi-encoder models to generate vector embeddings from catalogs, enabling fast indexing in elastic vector structures.

RESULT

Significant increase in click-through rates on the top search result provided and a major improvement in internal browsing satisfaction.

CASE #22

INDUSTRIAL PREDICTIVE MAINTENANCE WITH DEEP LEARNING AND IOT

Analytical Consulting Specialized in IT

CONTEXT (CHALLENGE)

Application of smart analytical approaches to anticipate severe structural failures in heavy shop-floor machinery.

SITUATION

Unforeseen downtime on industrial assembly lines due to mechanical breakdowns caused millions of dollars in losses per idle hour.

TASK

Predict anomalies in physical continuous telemetry sensors (vibration, temperature) before a breakdown occurs.

ACTION

Modeled LSTM neural networks focused on detecting anomalous patterns in time series from industrial IoT sensors.

RESULT

A proactive failure anticipation window of several days, reducing urgent corrective maintenance costs and safeguarding the production schedule.

CASE #23 ALGORITHMIC OPTIMIZATION OF COMPLEX DISTRIBUTION ROUTES

Analytical Consulting Specialized in IT

CONTEXT (CHALLENGE)

Use of operations research and combinatoric optimization algorithms focused on reducing fleet logistics movement costs.

SITUATION

Manual fleet delivery planning led to excessive redundant mileage, uncontrolled fuel spending, and missed contractual deadlines.

TASK

Optimize the routing network, calculating ideal paths while considering rigid time windows and cargo constraints.

ACTION

Implemented customized meta-heuristic algorithms in Python coupled with real-time geographic data.

RESULT

Drastic reduction in distances traveled and significantly optimized fuel costs across the integrated logistics operation.

CASE #24 FINOPS FOR GLOBAL AI PROJECTS (GPU/TPU OPTIMIZATION)

Leading Tech & AI Unicorn

CONTEXT (CHALLENGE)

Implementation of rigid governance and strategic financial monitoring over high-performance hardware consumption dedicated to large models.

SITUATION

Cloud infrastructure invoices scaled alarmingly due to model training instances being left idle or underutilized.

TASK

Ensure full efficiency in heavy AI computational processing consumption without throttling the delivery pace of data science fronts.

ACTION

Implemented specialized FinOps practices, finely monitored GPU memory usage via telemetry dashboards, and configured automated shutdown scheduling.

RESULT

Immediate structural reduction in monthly cloud computing costs for heavy models while maintaining PoC cadence and efficiency.

CASE #25

HIGH-PERFORMANCE ANALYTICS DEVELOPMENT WITH MASS PYSPARK

Major Marketplace & Retail Tech

CONTEXT (CHALLENGE)

Optimization of code and distributed processing structures aimed at accelerating heavy corporate analytical routines.

SITUATION

ETL jobs and routines built without optimization standards exceeded Apache Spark cluster memory limits, causing expensive re-processing.

TASK

Ensure fast and stable execution of batch data transformations without wasting computational infrastructure.

ACTION

Deeply refactored PySpark routines, eliminating inefficient network data movement (shuffling), applying broadcasting for small tables, and fine-tuning partitions.

RESULT

Execution time of analytical pipelines was drastically reduced, guaranteeing full processing stability in production.

CASE #26

HYPOTHESIS VALIDATION AND A/B TESTING CULTURE

Major Marketplace & Retail Tech

CONTEXT (CHALLENGE)

Installation of rigid scientific validation methodologies to test changes in digital products before definitive launch.

SITUATION

Changes to app layouts and algorithms were made based on personal opinions, without reliable statistical measurement of the real impact on sales.

TASK

Build a trustworthy continuous experimentation ecosystem anchored on rigid mathematical hypothesis testing concepts.

ACTION

Implemented an internal A/B testing pipeline integrated with automated calculations for statistical significance, ideal sample size, and statistical power in Python.

RESULT

Feature launches became strictly driven by solid empirical data, preventing financial losses from inefficient software changes.

CASE #27 MATRIX AUTOMATION OF IMAGES WITH PYTHON (PILLOW / LANCZOS)

MLGTT Consulting / MLGTT Consulting

CONTEXT (CHALLENGE)

Use of procedural programming logic and analytical matrix manipulation to optimize heavy graphic processes at speed.

SITUATION

An operational bottleneck existed in the manual processing and visual assembly of massive analytical ad mosaics, demanding days of manual work from the design team.

TASK

Automate rapid computational manipulation and mathematical resizing of thousands of batch images while maintaining high visual fidelity.

ACTION

Developed a robust utility class in pure Python focused on parallel image processing using specialized Lanczos resampling algorithms.

RESULT

Mosaics generated with surgical precision in minutes instead of days, completely eradicating manual editing efforts.

CASE #28 DESIGN OF ANALYTICAL ARCHITECTURES BASED ON DBT

Analytical Consulting Specialized in IT

CONTEXT (CHALLENGE)

Modeling stable data layers for business intelligence consumption through transformations controlled and documented via code.

SITUATION

Business rules for critical metrics (like revenue calculation) varied constantly from spreadsheet to spreadsheet, causing conflicting goal reports.

TASK

Centralize and control the mathematical governance of all essential corporate transformations and metrics within the organization.

ACTION

Structured data models completely using dbt coupled with a version control repository, integrating primary key tests and automated Data Lineage documentation.

RESULT

Established a single source of analytical truth in the company, ensuring absolute consistency and transparent governance over corporate figures.